



TARGET-90

JKBOSE CLASS-11TH

CHAPTER. NO. 8.
EQUILIBRIUM

One Shot

CHEMISTRY

Vishal Sir



JKBose Wallah

Topics To Be Covered

1 ... One Shot

2 ...

3 ...

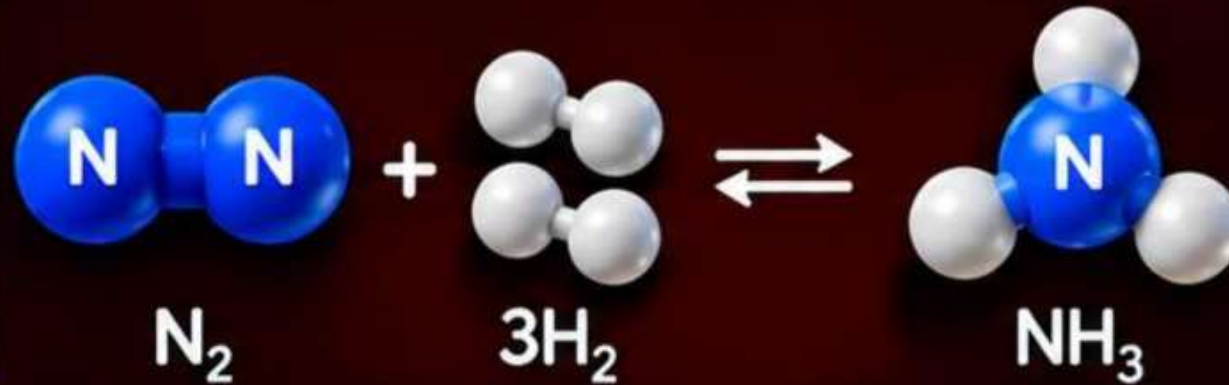
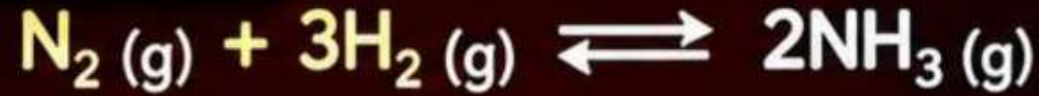
4 ...

EQUILIBRIUM

Definition

The state in which the rate of forward reaction becomes equal to the rate of backward reaction is called **chemical equilibrium**.

Example



Types of Equilibrium

1 Dynamic Equilibrium

- Both forward and backward reactions continue.
- Concentrations remain constant.
- Occurs in a closed system.

2 Static Equilibrium

- No further change occurs.
- Occurs in a closed system (mainly physical equilibrium).

Equilibrium Constant

For a general reaction:

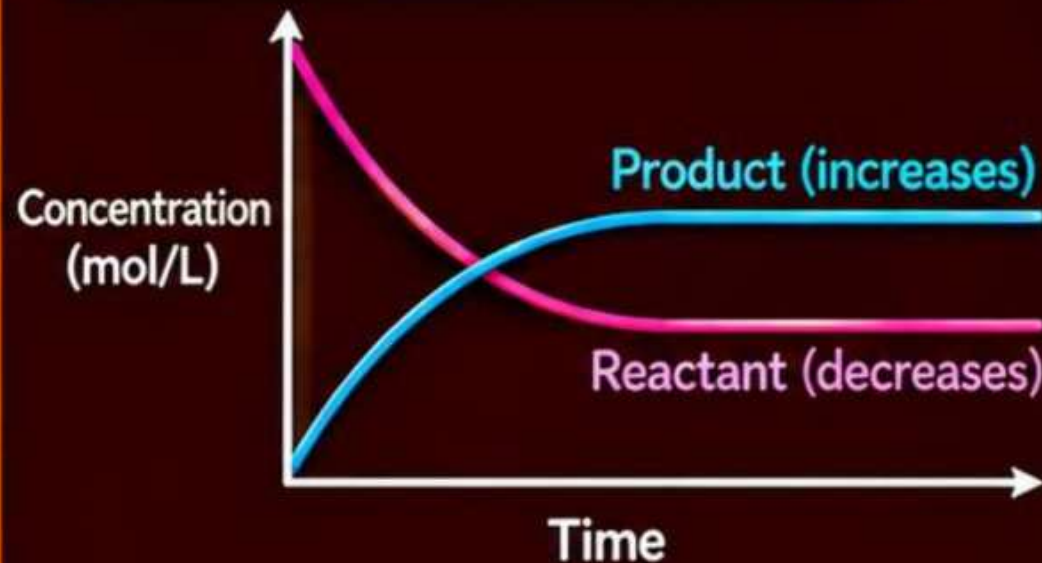


$$K_c = \frac{[\text{C}]^c [\text{D}]^d}{[\text{A}]^a [\text{B}]^b}$$

[] = molar concentration (at equilibrium)

K_c depends only on temperature.

Graphical Representation



Key Points

- At equilibrium, concentration of reactants and products remains constant.
- Rate of forward reaction = Rate of backward reaction.
- It can be achieved from either direction. (starting with reactants or products).
- Equilibrium can be shifted by changing concentration, pressure or temperature (**Le Chatelier's Principle**).

UNIT-VI: EQUILIBRIUM

07 MARKS

- Equilibrium in physical and chemical processes, dynamic nature of equilibrium, law of mass action, equilibrium constant, factors affecting equilibrium: Le Chatelier's principle equilibrium-ionization of acids and bases, strong and weak electrolytes, degree of ionization, Concept of pH. Hydrolysis of salts (elementary ideal, buffer solutions. Solubility product, common ion effect (with suitable examples).

Equilibrium and Its Classification



TYPES OF Physical Equilibrium.

Equilibrium – A state where temperature, pressure, and concentration of a system remain constant with time.

Types of Equilibrium:

1. **Physical Equilibrium:** Involves physical changes.



Examples: melting of ice, evaporation of water, dissolution of sugar.

2. **Chemical Equilibrium:** Involves chemical reactions.

Examples:



(1) Solid-Liquid Equilibrium

- If a piece of ice is placed in a thermos and allowed to melt, the weights of ice and liquid are measured.
- Now, if we close the thermos flask and maintain the temperature, we observe that the weights of ice and liquid water remain constant with time...
- This means: *there is equilibrium*



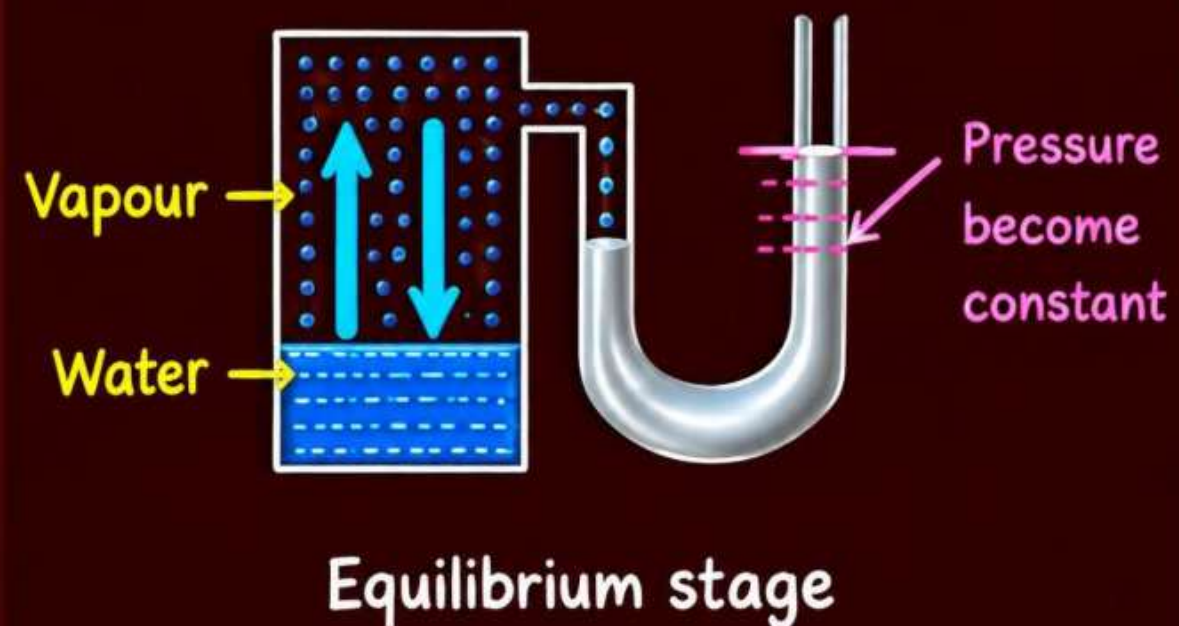
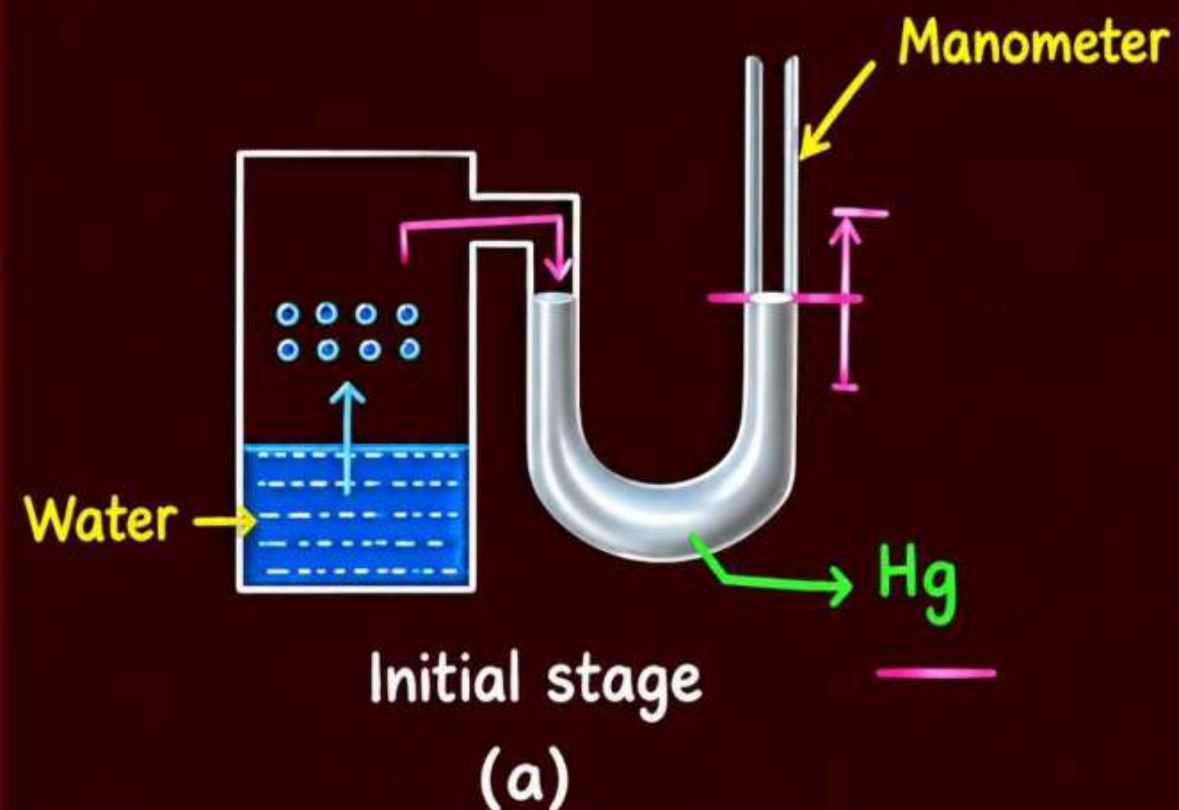
Examples of Physical Equilibrium

2. Liquid-Vapour Equilibrium

- When liquid water is placed in a closed container, it starts evaporating, increasing vapour pressure. This is observed as a rise in mercury level in a manometer.
- Eventually, a constant pressure is reached — indicating equilibrium.

At the molecular level:

- Evaporation produces vapour molecules.
- Vapour molecules collide with the liquid surface and condense.
- Initially, evaporation is faster than condensation, but over time both rates become equal.
- This balance maintains constant vapour pressure — the state of liquid-vapour equilibrium.



Examples of Physical Equilibrium



3. Solid-Vapour Equilibrium

- If we take iodine (I_2) in a closed container, it sublimates to form violet-coloured vapours.
- Initially, the intensity of violet colour increases as sublimation occurs rapidly.
- After some time, the intensity becomes constant — indicating equilibrium.



4. Solid-Solution Equilibrium

- If we take a glass of water and add NaCl, it dissolves to form an unsaturated solution.
- On adding more solute, crystals begin to appear — meaning no more solute can dissolve at that temperature.
- This is a saturated solution.
- At this stage:



Examples of Physical Equilibrium



5. Gas-Solution Equilibrium

- If we open a soda bottle, CO_2 gas fizzes out.
- This shows that the gas was dissolved under pressure.
- There is an equilibrium between:



According to Henry's Law:

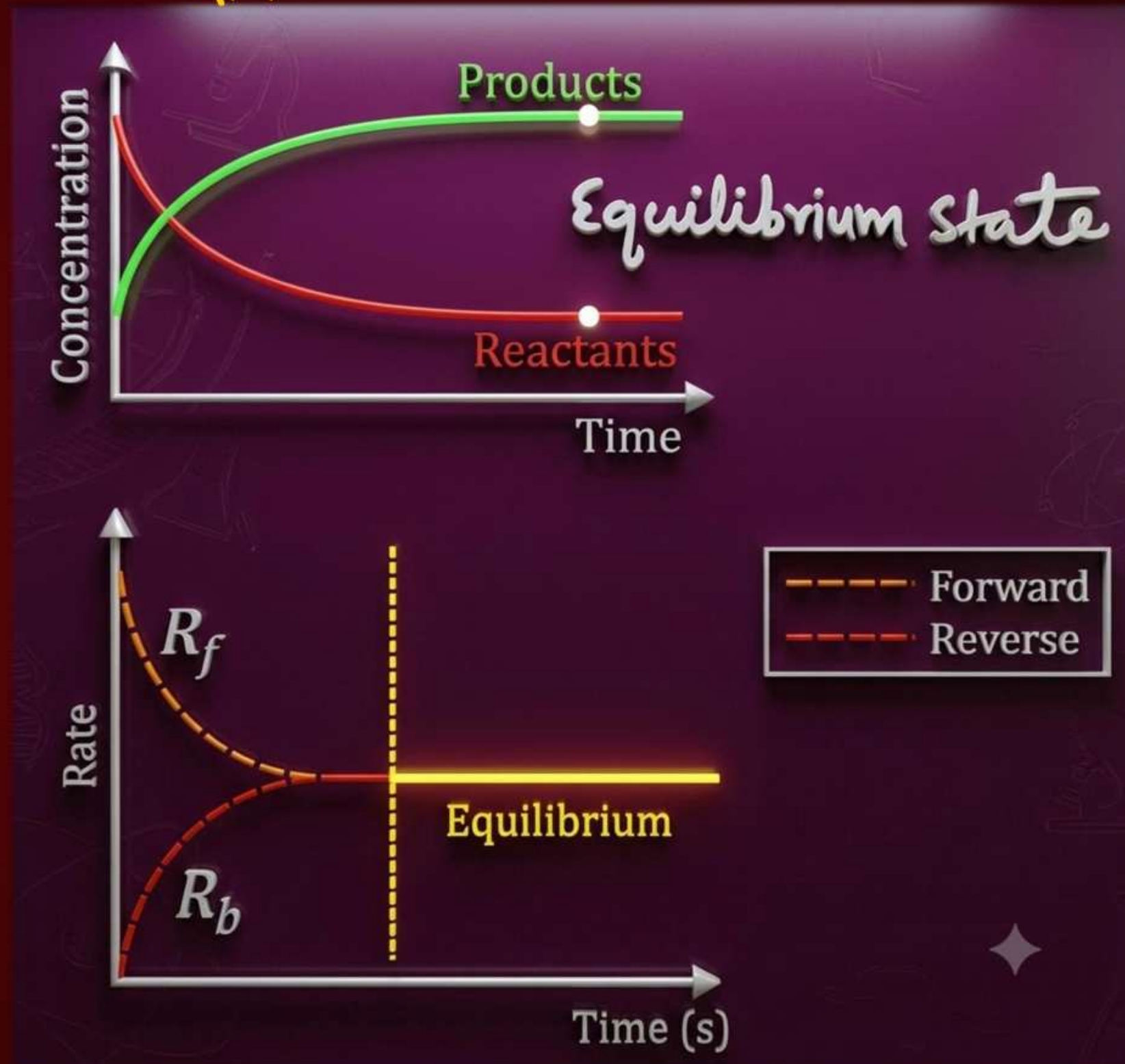
- "The mass of a gas dissolved in a liquid is directly proportional to the pressure of the gas above the liquid surface."

$$m_{(\text{gas})} \propto P_{(\text{gas})}$$



Chemical Equilibrium

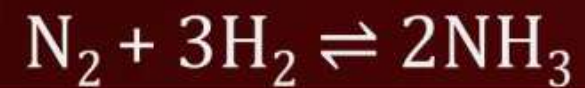
- In the case of reversible reactions, the stage at which the concentration of reactants and the concentration of products become constant is called chemical equilibrium.
- At this stage, the rate of the forward reaction becomes equal to the rate of the backward reaction.



Dynamic Nature of Chemical Equilibrium

- As we know, at equilibrium the **concentration, temperature, and pressure** remain constant.
- It seems that the chemical reaction has stopped or become static, but in reality,
- the **reaction continues to occur in both directions.**
- Hence, it is said to be **dynamic in nature.**

Example Haber's Process:



- At equilibrium, the concentration of all reactants and products remains constant at a particular temperature.
- If we replace H_2 with its isotope D_2 :
- $\text{N}_2 + 3\text{D}_2 \rightleftharpoons 2\text{ND}_3$
- If we mix H_2 and D_2 together, we find at equilibrium a **mixture of isotopic compounds.**
 $\text{H}_2, \text{D}_2, \text{HD}, \text{NH}_3, \text{ND}_3, \text{NHD}_2, \text{NH}_2\text{D}.$
- This proves that **the reaction is still proceeding, i.e., chemical equilibrium is dynamic in nature.**

Characteristics of Chemical Equilibrium

1. At constant temperature, pressure, concentration, colour, and density remain constant.
2. Equilibrium can be reached from both directions — forward or backward.
- ★ 3. A catalyst doesn't change equilibrium position, it only helps to reach it faster.
4. Equilibrium is dynamic in nature — both reactions continue with equal rate.
5. It remains unchanged until external conditions like temperature or concentration are altered.

Laws of Mass Action

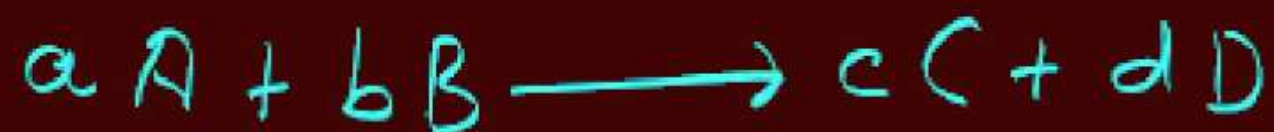
(Law of Chemical Equilibrium)



Given by: Guldberg and Waage

It states that:

- "The rate of a chemical reaction is directly proportional to the active mass (concentration) of the reactants, each raised to the power of its stoichiometric coefficient."



$$\text{Rate} \propto [A]^a [B]^b$$

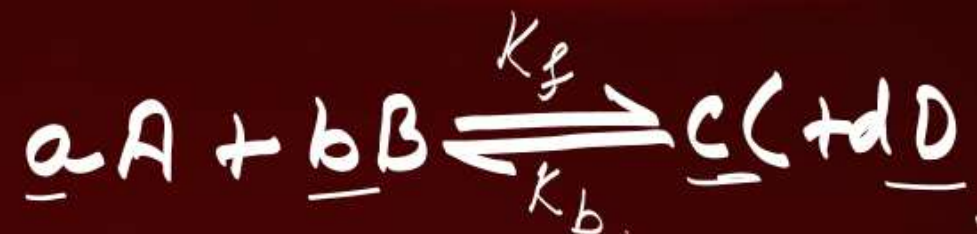
$$\text{Rate} = K [A]^a [B]^b$$

↓
Rate constant.

Laws of Mass Action



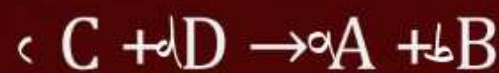
- Let us consider a simple reversible reaction,



According to law of mass action,

Rate of forward reaction $\propto [A]^a [B]^b = k_f [A]^a [B]^b$

Similarly, the backward reaction is:



Rate of backward reaction $\propto [C]^c [D]^d = k_b [C]^c [D]^d$

At equilibrium $R_f = R_b$.



$$k_f [A]^a [B]^b = k_b [C]^c [D]^d$$

$$\frac{k_f}{k_b} = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

$$K_{eq} = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

- The product of the molar concentrations of the products, each raised to the power equal to its stoichiometric coefficient divided by the product of the molar concentrations of the reactants, each raised to the power equal to its stoichiometric coefficient is constant at constant temperature and is called Equilibrium constant. (Law of Chemical Equilibrium).

Relationship Between K_p and K_c



K_p = equilibrium constant in terms of partial pressure, expressed in mole fraction.

K_c = equilibrium constant in terms of concentration at a given pressure.

$$\frac{10^7}{10^8} = 10^{7-8}$$

Consider a gaseous Rxn.



According to Law of chemical equilibrium,

$$K_p = \frac{p_C^c p_D^d}{p_A^a p_B^b}$$

According to ideal gas equation, $pV = nRT$.

$$\text{or } p = \frac{n}{V} RT$$

$$\text{or } \boxed{P = CRT}$$

$$p_A^a = (C_A^* RT)^a$$

$$p_B^b = (C_B^* RT)^b$$

$$p_C^c = (C_C^* RT)^c$$

$$p_D^d = (C_D^* RT)^d$$

$$\begin{aligned} K_p &= \frac{(C_C RT)^c (C_D RT)^d}{(C_A RT)^a (C_B RT)^b} = \frac{C_C^c (RT)^c C_D^d (RT)^d}{C_A^a (RT)^a C_B^b (RT)^b} \\ &= \frac{C_C^c C_D^d (RT)^{c+d}}{C_A^a C_B^b (RT)^{a+b}} = \frac{C_C^c C_D^d}{C_A^a C_B^b} (RT)^{c+d-(a+b)} \\ &= \frac{C_C^c C_D^d}{C_A^a C_B^b} (RT)^{\Delta n_g} \quad (c+d)-(a+b) = \Delta n_g \\ &= K_c (RT)^{\Delta n_g} \end{aligned}$$

$$\boxed{K_p = K_c (RT)^{\Delta n_g}}$$

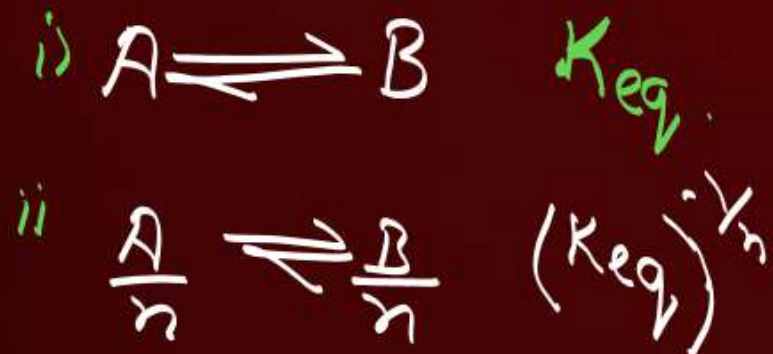
$$\boxed{\frac{K_p}{K_c} = (RT)^{\Delta n_g}}$$

Characteristics of Equilibrium Constant



1. Equilibrium constant (K) does not depend on the initial concentration of reactants.
It depends only on temperature.

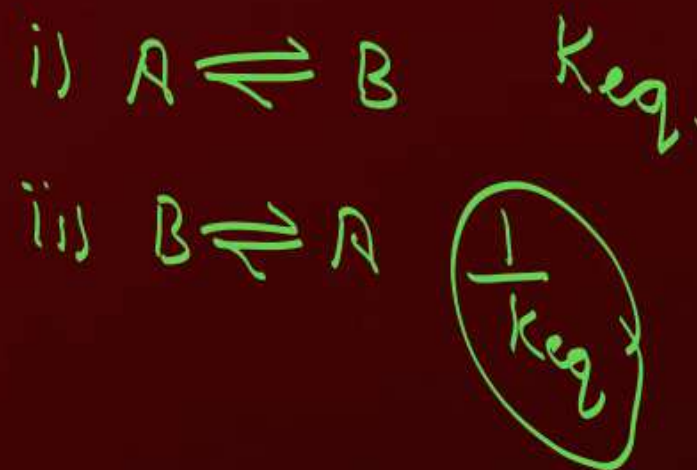
2. If we divide the chemical equation by 'n', then the new equilibrium constant is:



3. If we multiply the chemical equation by 'n', then the new equilibrium constant is:



4. If we reverse the chemical equation, then the new equilibrium constant becomes the reciprocal of the original:



Characteristics of Equilibrium Constant



5. If we add two equilibrium equations, then the overall equilibrium constant is the product of the two:



$$K_{eq} = K_1 \times K_2$$

6. If we subtract one equilibrium equation from another, then the resultant equilibrium constant is the ratio of the two:



$$K_{eq} = \frac{K_1}{K_2}$$

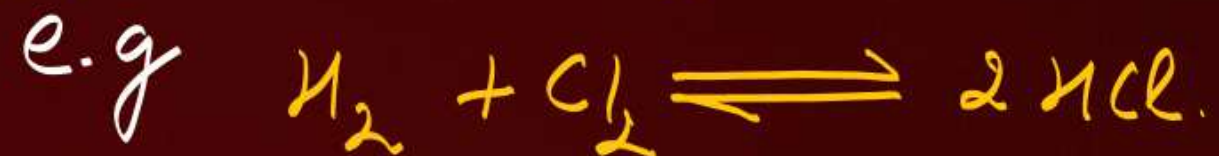
Le Chatelier Principle

(Most Imp)



- Le-Chatelier and Braun (1884), French chemists.
- If a system in equilibrium is subjected to a change of concentration, temperature or pressure, then equilibrium shifts in a direction that tends to undo the effect of the change imposed.

(1) Effect of Change of Concentration



If we add any one of reactant to the rxn, i.e. increase in conc. of reactant

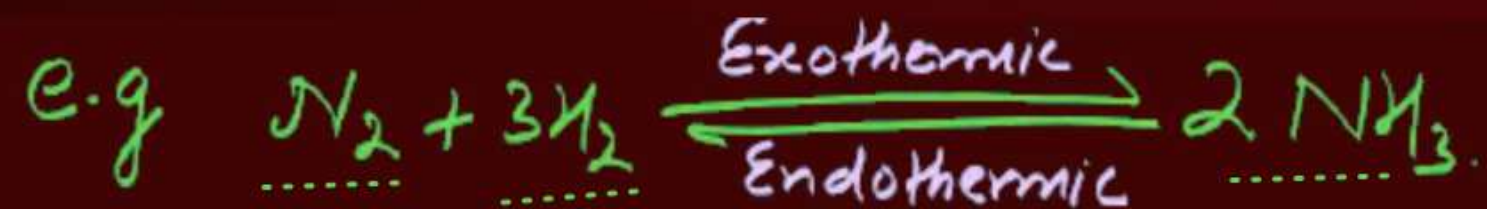
- Equilibrium will shift in that direction where reactant concd. decrease.
i.e. forward direction

& Vice Versa.

Le Chatelier Principle



(2) Effect of Change of Temperature



If $T \uparrow$ (heat $\uparrow$), Endothermic Reaction will be favoured.

If $T \downarrow$, Exothermic Reaction will favour.

(3) Effect of Change of Pressure

We know that (no. of mole of gas & P_{gas}).



• If $P \uparrow$ (no. of moles of gas $\uparrow$), Equilibrium will shift in a direction having less no. of moles.

If $P \downarrow$ (no. of moles of gas $\downarrow$), Equilibrium will shift in a direction have more no. of moles.

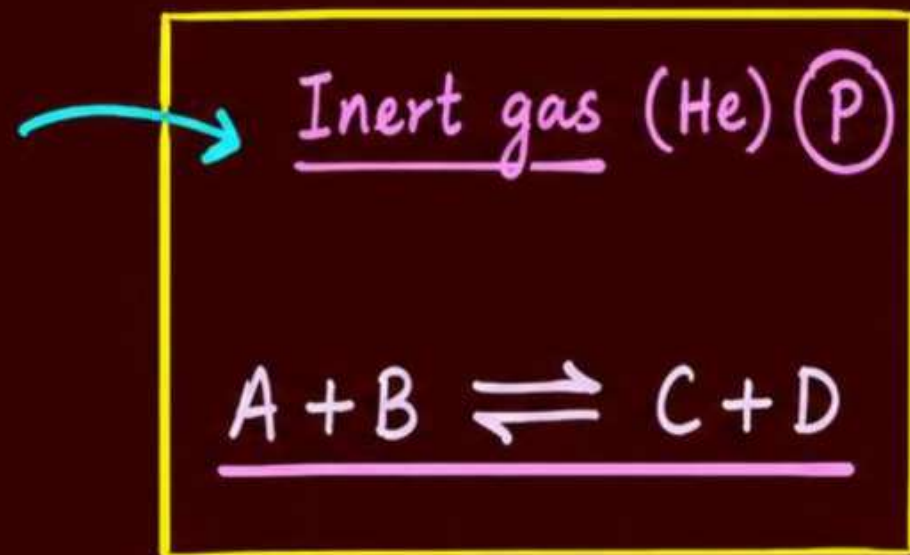
Le Chatelier Principle



① Effect of addition of inert gas

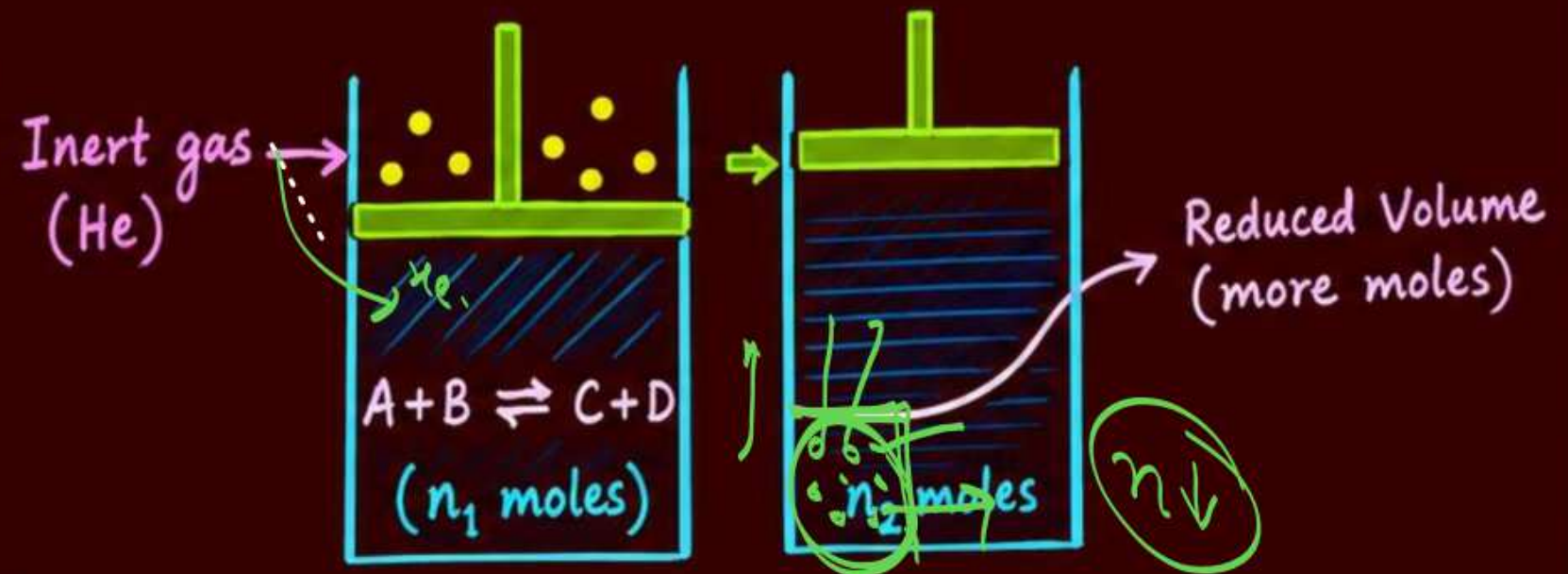
a) At constant volume.

When we add inert gas, partial pressure of reactant ~~and~~ ^{product} does not change but does not anything about reactant to product.
→ Equilibrium remain unaffected.



b) At constant Pressure.

Addition of inert gas, will increase the volume of system increases which results no. of moles decreases, therefore equilibrium will shift in a direction having more no. moles.



Industrial Application of Le-Chatelier



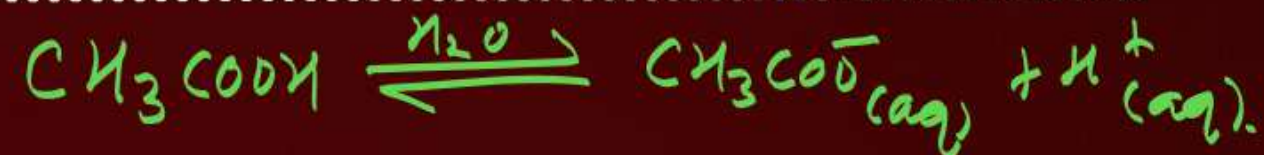
- (a) In the formation of **Ammonia** (Haber process).
- (b) In the formation of **Sulphuric acid** (Contact Process)
(H_2SO_4).

Ionic Equilibrium



When equilibrium exists between ions in a solution, it is called ionic equilibrium.

Example:



Not all substances ionize in water. Based on their ability to dissociate in water, Michael Faraday classified substances into two types:

(a) Non-electrolytes (b) Electrolytes

(a) Non-electrolytes:

Substances that do not dissociate into ions in water.

They do not conduct electricity.

Examples: Urea, Glucose, Fructose

(b) Electrolytes:

Substances that dissociate into ions in water.

They conduct electricity.

Examples: All acids, bases, and salts like HCl, NaOH, KOH, NaCl, etc.

Ionic Equilibrium



Based on the degree of dissociation (α), electrolytes are of two types:

1. Strong electrolytes
2. Weak electrolyte

Degree of Dissociation (α):
$$\frac{\text{No. of moles dissociated}}{\text{Total no. of moles present}}$$

Types of Electrolytes:

1. Weak Electrolytes:

Dissociate partially or to a small extent in water.

Examples: CH_3COOH , H_3BO_3 , NH_4OH , H_2CO_3

2. Strong Electrolytes:

Dissociate completely in water.

Examples: HCl , HI , NaCl , KCl , NaOH , KOH

Concept of Acids and Bases

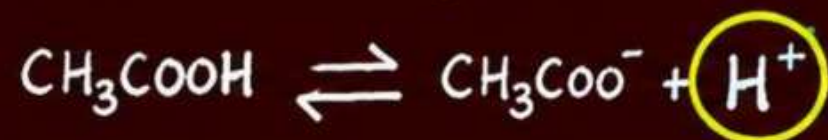
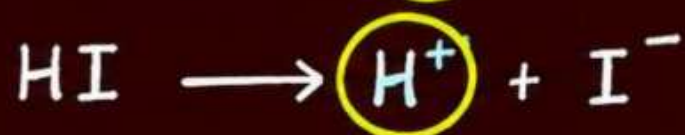
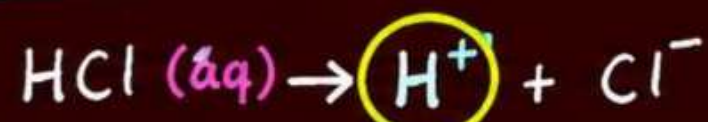


1. Arrhenius Concept

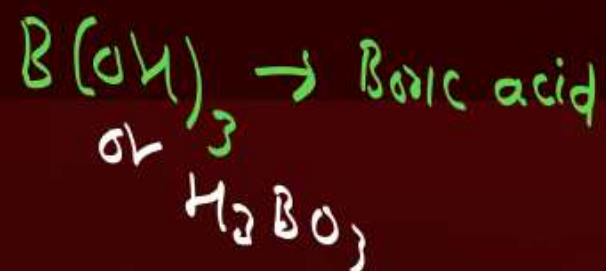
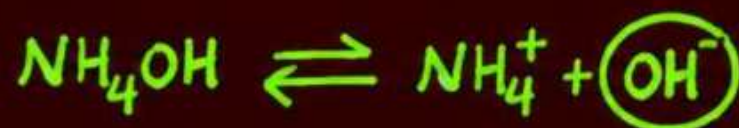
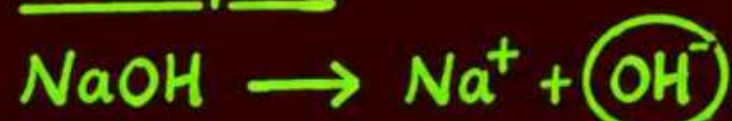
Acids: Substances that produce H^+ (protons) in water.

Bases: Substances that produce OH^- ions in water.

Example: Acid.



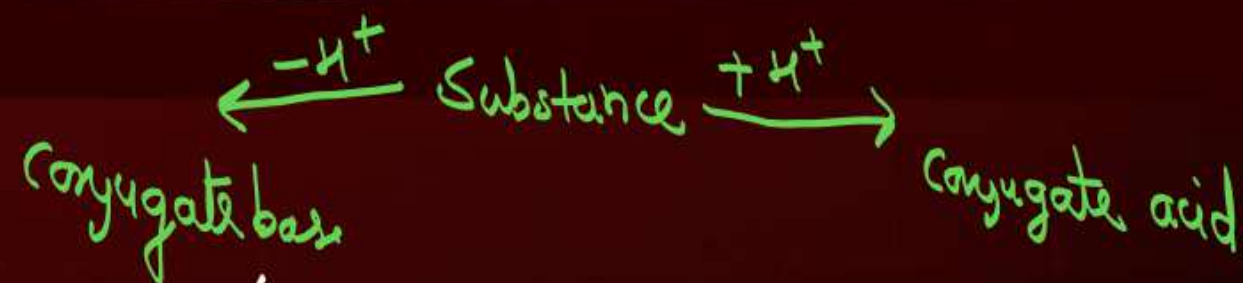
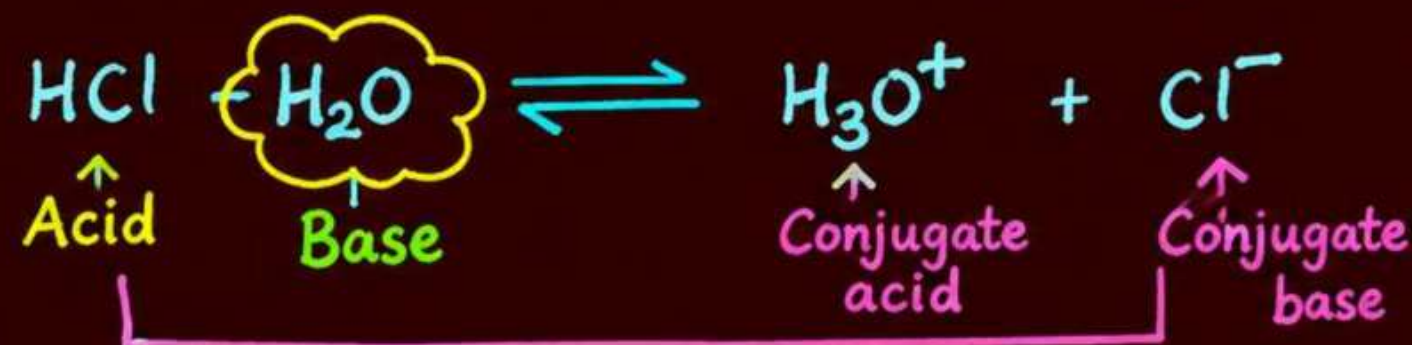
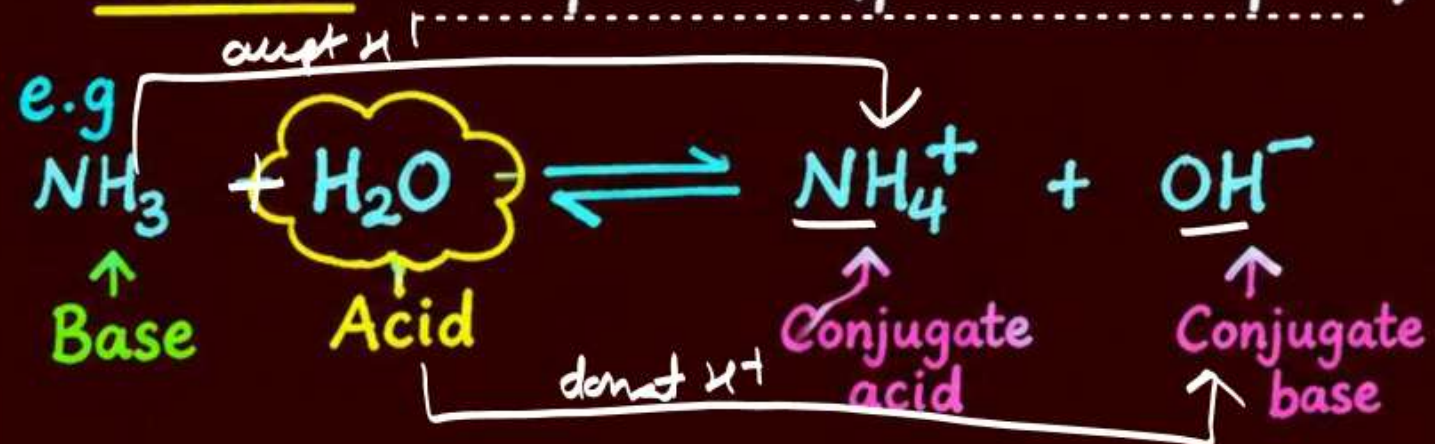
Example: Base



2. Brønsted-Lowry Concept

Acids: Donates H^+ (proton donor)

Bases: Accepts H^+ (proton acceptor)

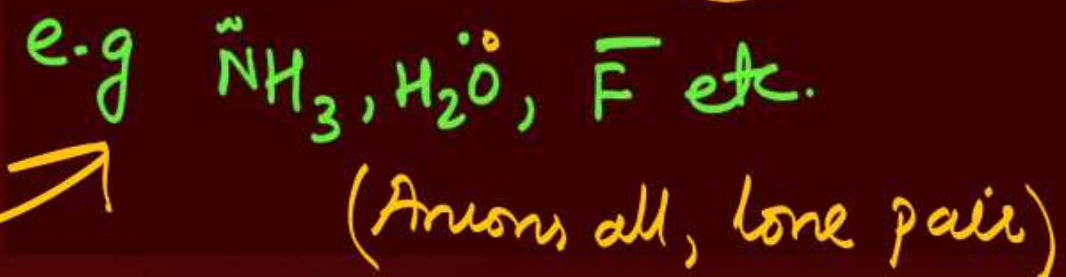
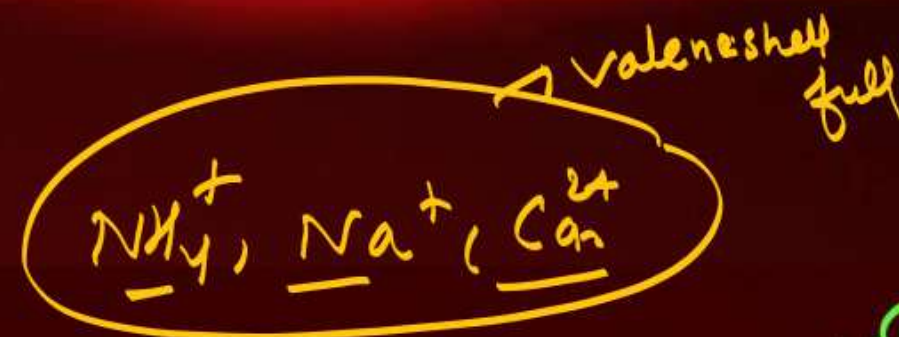


Concept of Acids and Bases



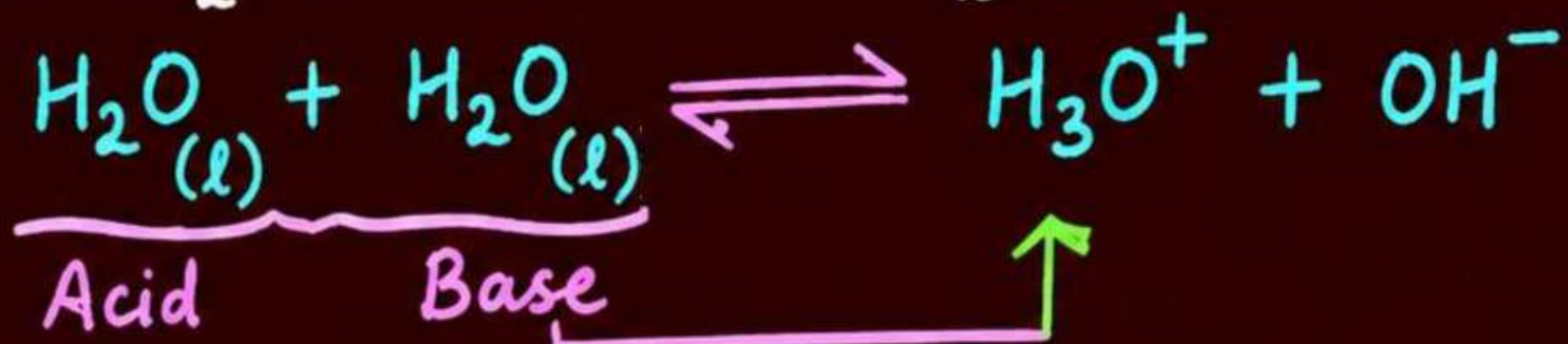
Lewis Acids and Bases

- According to this concept:
- Lewis acid $\rightarrow$ accepts a lone pair of electrons.
- Lewis base $\rightarrow$ donates a lone pair of electrons.



Ionic Product of Water (K_w)

When H_2O is dissolved in H_2O , then.



According to law of chemical equilibrium,

$$K_{eq} = \frac{[H_3O^+][OH^-]}{[H_2O]^2 \text{ or } [H_2O][H_2O]}$$

Note → Conc. of pure solid & pure liquid is taken as unity or constant.

$$\underline{K_{eq}} \times [H_2O]^2 = [H_3O^+][OH^-]$$

$$K_w = [H_3O^+][OH^-]$$

At 298 K.



$$[H_3O^+][OH^-] = [10^{-7}][10^{-7}]M$$

$$K_w = 10^{-7} \times 10^{-7}$$

$$K_w = 10^{-14}$$

Concept of pH

H^+ or H_3O^+ ek hi baad hai.



- Sorensen in 1909, suggested a new term for expressing the concentration of hydrogen ions known as pH or pH scale.
- The pH of a solution is defined as the negative logarithm of the H_3O^+ ion concentration in moles per litre.

Mathematically, it may be expressed as :

$$p^H = -\log[H^+]$$

$$p^H = \frac{1}{\log[H^+]}$$

At 298 K, $[H^+] = 10^{-7}$

$$p^H = -\log 10^{-7} \quad (\log x^a = a \log x)$$

$$p^H = 7 \log 10 \quad (\log 10 = 1)$$

$$p^H = 7 \times 1$$

$$p^H = 7 \quad \text{i.e. neutral.}$$

↓
Hydronium ion
Neutral

Acid 7 Basic 14

If $p^H > 7$, then solution is basic.

$p^H < 7$, then solution is acidic.

$p^H = 7$, then solution is neutral.

Relationship b/w p^H & p^{OH}

Just take, $p^H = -\log [H^+]$.

$$p^{OH} = -\log [OH^-].$$

According to ionic product of water,

$$K_w = [H^+][OH^-].$$

At 298 K,

$$10^{-14} = [H^+][OH^-].$$

Taking on b/s,

$$\log 10^{-14} = \log [H^+][OH^-] \quad \left. \vphantom{\log 10^{-14} = \log [H^+][OH^-]} \right\} \log a \times b = \log a + \log b$$

$$\log 10^{-14} = \log [H^+] + \log [OH^-]$$

$$\log 10^{-14} = \log [H^+] + \log [OH^-]$$

$$-14 \log 10 = \log [H^+] + \log [OH^-].$$

$$-14 = -p^H - p^{OH}.$$

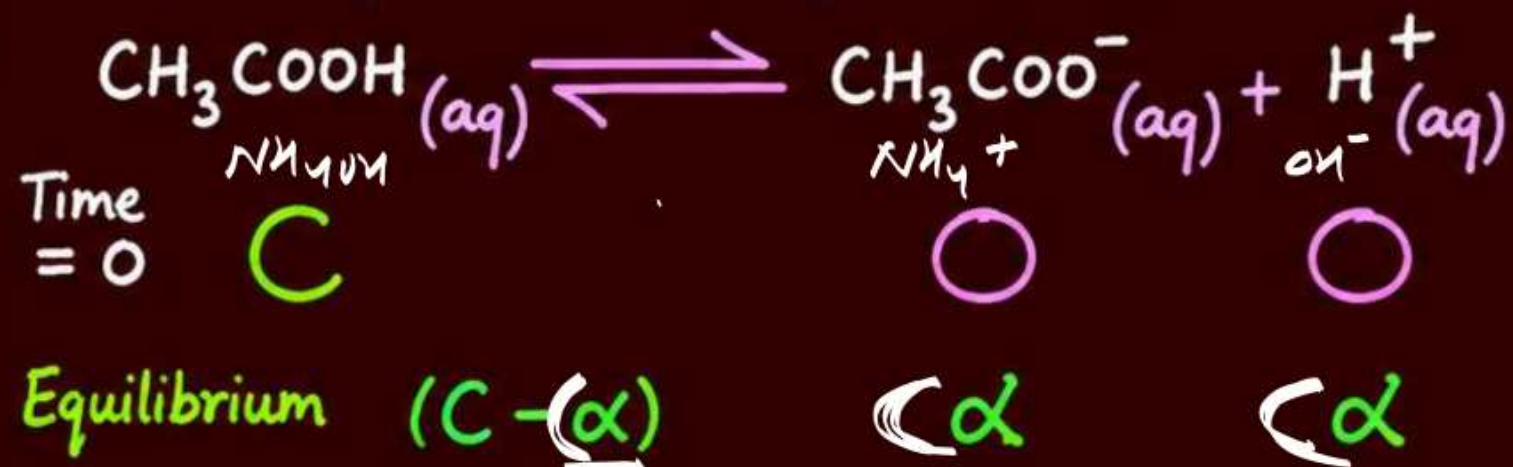
$$14 = -(p^H + p^{OH}).$$

$$\underline{p^H + p^{OH} = 14.}$$

Degree of Dissociation of Weak Acids and Weak Bases



As we know, weak acid does not dissociate completely in water e.g. CH_3COOH .



According to law of chem. equilibrium

$$K_a = \frac{[\text{CH}_3\text{COO}^-][\text{H}^+]}{[\text{CH}_3\text{COOH}]}$$

$$K_a = \frac{C\alpha \cdot C\alpha}{C - \alpha} = \frac{C\alpha^2}{1 - \alpha}$$

$$K_b = \frac{C\alpha^2}{(1 - \alpha)} = \frac{C\alpha^2}{1 - \alpha}$$

(α max value = 1)

For a weak Acid $1 \gg \alpha$, can be neglected as a denominator.

$$K_a = C\alpha^2$$

or

$$\alpha^2 = \frac{K_a}{C}$$

$$\alpha = \sqrt{\frac{K_a}{C}}$$

$$\alpha \propto \frac{1}{\sqrt{C}}$$

Salts Hydrolysis

When salt is dissolved in water produces Acid & Base
this is called salt hydrolysis.



opposite Rxn i.e. $\text{Acid} + \text{Base} \rightarrow \text{Salt} + \text{H}_2\text{O}$
i.e. neutralisation Rxn.



Types of salt

I. Salts of strong acids and strong bases:

Examples are:



As an illustration, let us discuss the hydrolysis of NaCl. We may write :



II. Salts of weak acids and strong bases:

Examples are:



As an illustration, the hydrolysis of sodium acetate (CH_3COONa) may be represented as follows:



Salts Hydrolysis

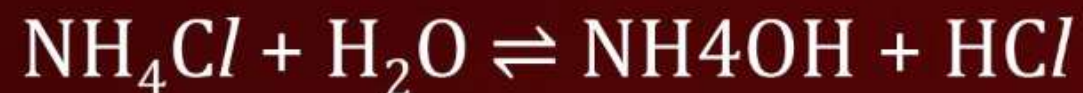


III. Salts of strong acids and weak bases:

Examples are:

NH_4Cl , CuSO_4 , NH_4NO_3 , AlCl_3 , CaCl_2 etc.

As an illustration, the hydrolysis of NH_4Cl may be represented as follows:

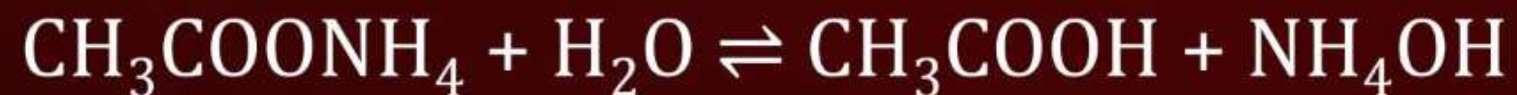


IV. Salts of weak acids and weak bases:

Examples are:

$\text{CH}_3\text{COONH}_4$, $(\text{NH}_4)_2\text{CO}_3$, AlPO_4 etc.

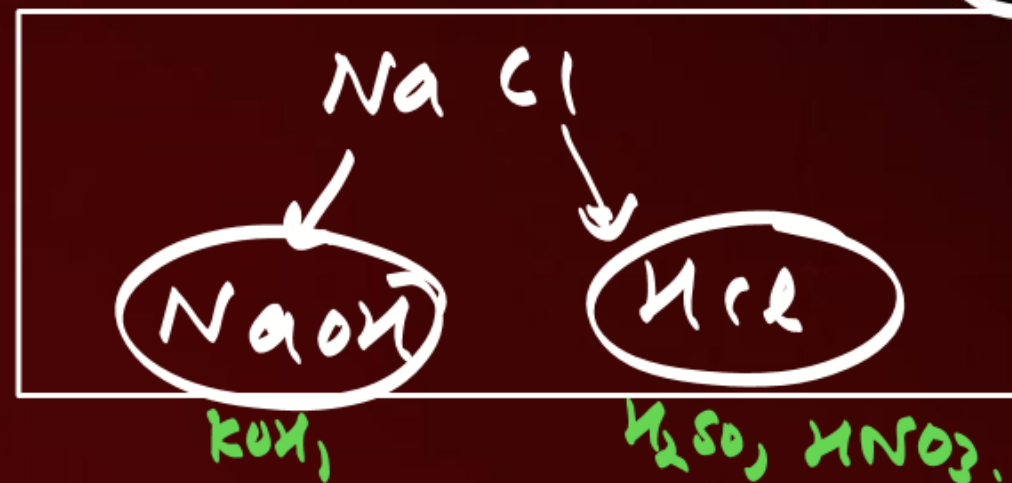
As an illustration, the hydrolysis of ammonium acetate may be represented as follows:



Types

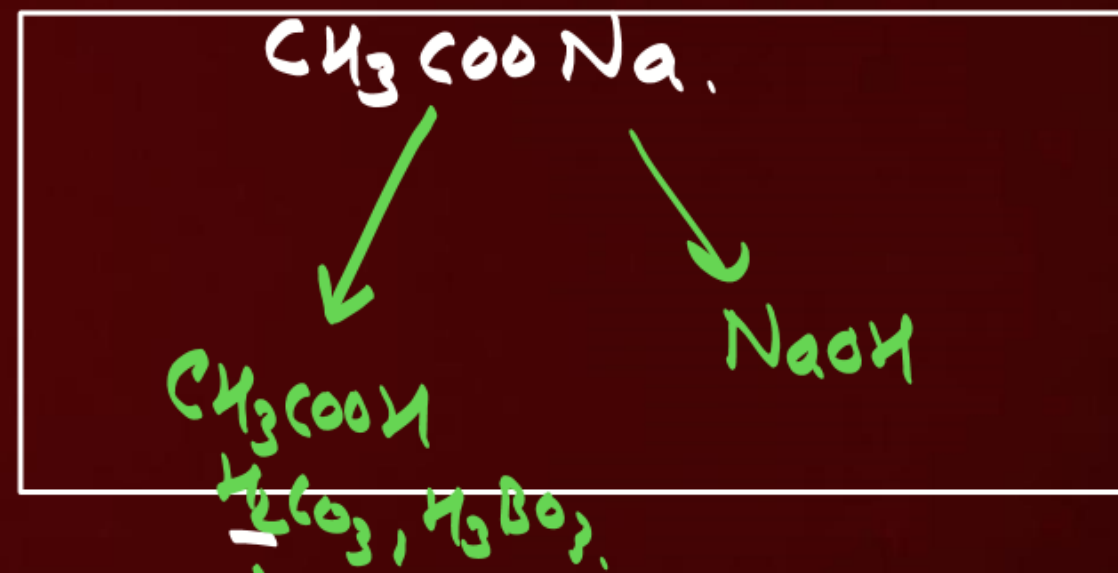
1) Salt of strong acid & strong base.

e.g. NaCl , Na_2SO_4 , KNO_3 , $\text{pH} = \text{neutral}$



2) Salt of Weak Acid & strong base.

CH_3COONa , Na_2CO_3 , K_2CO_3 etc



3) Salt of strong Acid & Weak Base.

e.g. NH_4Cl , CuSO_4 , $\text{Al}_2(\text{SO}_4)_3$ etc

4) Salt of Weak Acid & Weak base

$\text{CH}_3\text{COO}^-\text{NH}_4^+$, CuCO_3 ,

Weak Base is NH_4OH , $\text{Cu}(\text{OH})_2$, $\text{Al}(\text{OH})_3$.

Common Ion Effect



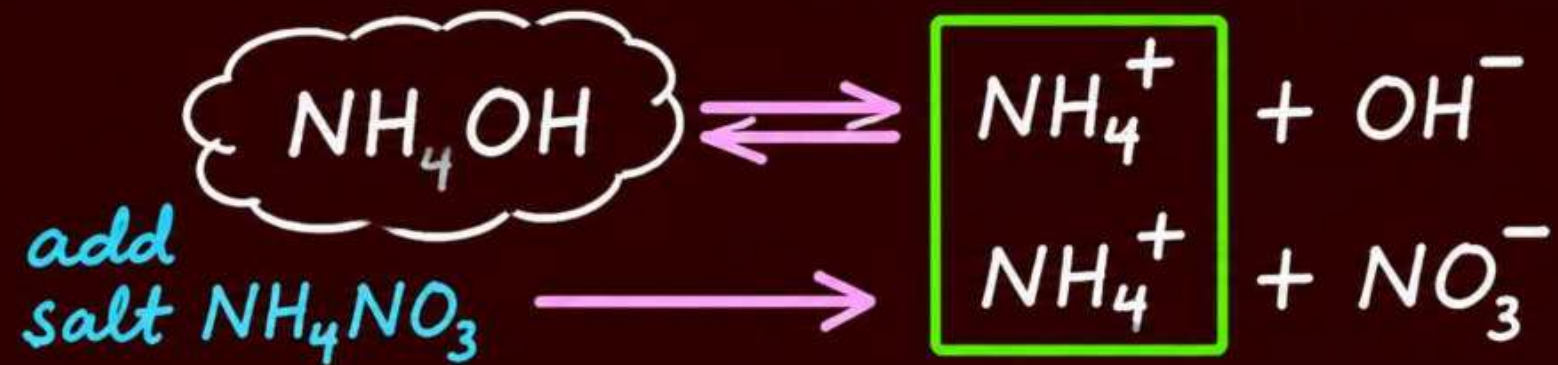
- When a salt containing a common ion is added to an ionic equilibrium, the concentration of that ion increases. *in product side*.
- According to **Le-Chatelier's Principle**, the equilibrium shifts in the **backward direction**, reducing the dissociation of the weak electrolyte.

Example 1:

- For $\text{CH}_3\text{COOH} \rightleftharpoons \text{CH}_3\text{COO}^- + \text{H}^+$
- Adding CH_3COONa increases CH_3COO^- concentration, so the dissociation of CH_3COOH decreases. *in product, hence rxn will shift in backside*

Example 2:

- For $\text{NH}_4\text{OH} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$ *in product*
- Adding NH_4Cl increases NH_4^+ concentration, so the dissociation of NH_4OH decreases.



Solution NH_4^+ conc. $\uparrow$ i.e. Equilibrium backward shift i.e. dissociation occurs.

Buffer Solution

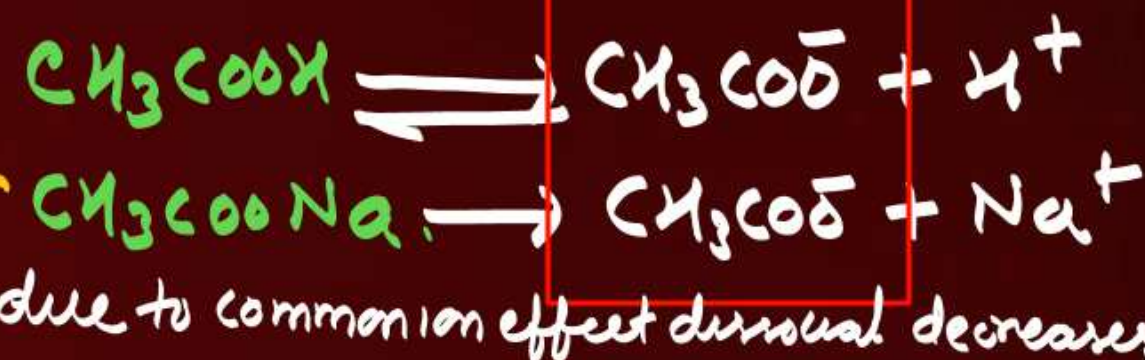


- A **buffer solution** is one whose **pH remains nearly constant** on adding a small amount of acid or base.
- This ability to resist pH change is called **buffer action**.

Types of Buffers:

1. Acidic Buffer:

- Mixture of a **weak acid** and its **conjugate base** (salt of weak acid and strong base).
- **Example:** $\text{CH}_3\text{COOH} + \text{CH}_3\text{COONa}$
- **On adding acid (H^+):** $\text{CH}_3\text{COO}^- + \text{H}^+ \rightarrow \text{CH}_3\text{COOH}$, hence $[\text{H}^+]$ does not $\uparrow$ se.
- **On adding base (OH^-):** $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$, hence $[\text{OH}^-]$ does not $\uparrow$ se.
- Hence, pH remains nearly constant.



Buffer Solution



2 Basic Buffer:

- Mixture of a **weak base** and its **conjugate acid** (salt of weak base and strong acid).
- **Example:** $\text{NH}_4\text{OH} + \text{NH}_4\text{Cl}$
- $\text{NH}_4\text{OH} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$
- $\text{NH}_4\text{Cl} \rightarrow \text{NH}_4^+ + \text{Cl}^-$, due to common ion effect, dissociation of NH_4OH ↓ se.
- **On adding acid (H^+):** $\text{H}^+ + \text{OH}^- \rightarrow \text{H}_2\text{O}$, $[\text{H}^+]$ does not ↑ se
- **On adding base (OH^-):** $\text{NH}_4^+ + \text{OH}^- \rightarrow \text{NH}_4\text{OH}$, $[\text{OH}^-]$ does not ↑ se.
- Thus, pH does not change.

$$\text{Buffer capacity } (\phi) = \frac{\text{Number of moles of acid or base added to 1 L of buffer}}{\text{Change in pH}}$$



Solubility Product (K_{sp})

Some salts like AgBr , AgCl , PbCl_2 etc dissolve in a water to a very small extent, these salts are called sparingly soluble salt.

Dissociation of sparingly soluble salt like AgBr can be written as,



Law of Chemical Equilibrium

$$K = \frac{[\text{Ag}^+][\text{Br}^-]}{[\text{AgBr}]}$$

AgBr is pure solid, its conc. remains constant.

$$K \times [\text{AgBr}] = [\text{Ag}^+][\text{Br}^-]$$



$$K_{sp} = [\text{Ag}^+][\text{Br}^-]$$

K_{sp} = Solubility Product.

It is defined as product of concentration of ions of sparingly soluble salt.

Their molar solubility can also be calculated because their concentration is regarded as their solubility (S).

$$K_{sp} = [\text{Ag}^+][\text{Br}^-]$$

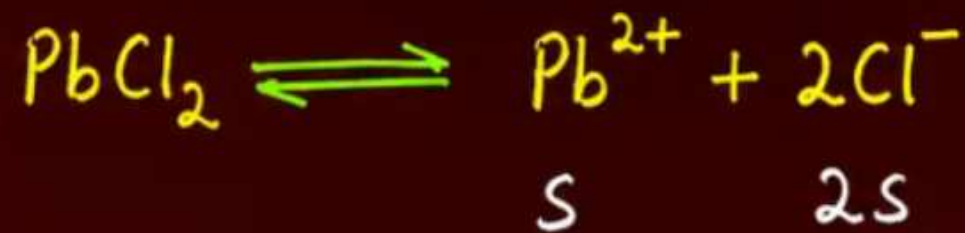
$$K_{sp} = S \times S$$

$$\boxed{K_{sp} = S^2} \quad \text{or } S = \sqrt{K_{sp}}$$

Find solubility. of following salts



H.W.



$$K_{sp} = [\text{Pb}^{2+}] [\text{Cl}^-]^2$$

$$K_{sp} = S (2S)^2$$

$$K_{sp} = S \times 2^2 S^2$$

$$= S \times 4S^2$$

$$K_{sp} = 4S^3$$

$$\frac{K_{sp}}{4} = S^3$$

$$S = \left(\frac{K_{sp}}{4} \right)^{1/3}$$



$$K_{sp} = [\text{Ba}^{2+}] [\text{SO}_4^{2-}]$$

$$K_{sp} = S \times S$$

$$K_{sp} = S^2$$

$$S = \sqrt{K_{sp}}$$



$$K_{sp} = [\text{Ca}^{2+}]^3 [\text{PO}_4^{3-}]^2$$

$$= (3S)^3 (2S)^2$$

$$= 3^3 S^3 \times 2^2 S^2 = 27 \times 4 S^5$$

$$= 108 S^5$$

$$K_{sp} = 108 S^5$$

$$S^5 = \frac{K_{sp}}{108}$$

$$S = \left(\frac{K_{sp}}{108} \right)^{1/5}$$

HOMEWORK



$$K_{sp} = [\text{Mg}^{2+}] [\text{OH}^-]^2$$

$$= s \times (2s)^2$$

$$= s \times 4s^2$$

$$K_{sp} = 4s^3$$

$$s^3 = \frac{K_{sp}}{4}$$

$$s = \left(\frac{K_{sp}}{4} \right)^{1/3}$$



$$K_{sp} = [\text{Ca}^{2+}]^3 [\text{PO}_4^{3-}]^2$$

$$= (3s)^3 (2s)^2$$

$$= 3^3 \times s^3 \times 2^2 \times s^2$$

$$= 27 \times 4 \times s^5$$

$$K_{sp} = 108 s^5$$

$$s^5 = \frac{K_{sp}}{108}$$

$$s = \left(\frac{K_{sp}}{108} \right)^{1/5}$$

QUESTION



Define pH. What is the pH of a solution whose $[H_3O^+] = 2 \times 10^{-5} M$?

[2024]

Sol $\Rightarrow$

$$\log 1 = 0$$

$$2 =$$

$$3 =$$

$$4 =$$

$$5 =$$

$$6 =$$

$$7 =$$

$$8 =$$

$$9 =$$

$$\log 10 = 1$$

$$pH = -\log [H_3O^+].$$

$$= -\log [2 \times 10^{-5}].$$

$$= -\left[\log 2 + \log 10^{-5} \right]$$

$$= -\left[0.301 - 5 \log 10 \right]$$

$$= -\left[0.301 - 5(1) \right]$$

$$= -\left[0.301 - 5 \right]$$

$$= -\left[-4.7 \right] = 4.7 \text{ Ans.}$$

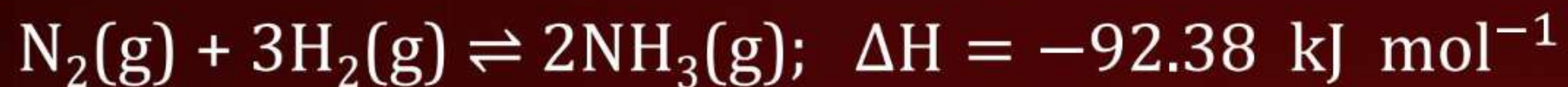
$$\left\{ \log a \times b = \log a + \log b \right\}$$

$$\log 10 = 1$$

QUESTION



State Le-Chatelier's principle. Apply this principle on the following reaction in terms of concentration, temperature, and pressure to obtain maximum yield of ammonia:



[2024]

QUESTION



Explain common ion effect with a suitable example.

[2024]

QUESTION



What are strong and weak electrolytes? Derive an expression for the calculation of the degree of ionisation of a weak electrolyte. **[2025]**

QUESTION



State Le Chatelier's principle. Give one example of its application in chemical industries.

[2025]

QUESTION



What are buffer solutions? How can you say that buffers are used as an important industrial processes? [2025]

Applications

- 1). Maintains blood pH (By buffer action of Carbonic acid)
- 2). Industrial application :→
 - Maintains pH of dye
 - drugs
 - Paints
 - paper etc.
- 3). Agriculture :→
 - To maintain the pH of soil.

Thank
You